

For the use only of a Registered Medical Practitioner or a Hospital or a Laboratory

Mactrivir

Efavirenz 600 mg , Emtricitabine 200 mg & Tenofovir Disoproxil Fumarate 300 mg Tablets

COMPOSITION

Each film coated tablet contains:

Efavirenz USP....600 mg

Emtricitabine....200 mg

Tenofovir Disoproxil Fumarate..... 300 mg equivalent to Tenofovir Disoproxil.....245 mg

PRODUCT DESCRIPTION

Pink Capsule shaped biconvex film coated tablets debossed "CL 81" on one side and plain on other side.

PHARMACOLOGICAL ACTION:

Pharmacokinetics:

Absorption

In HIV infected patients, peak efavirenz plasma concentrations were attained by 5 hours and steady- state concentrations reached in 6 to 7 days. Emtricitabine is rapidly absorbed with peak plasma concentrations occurring at 1 to 2 hours post-dose. Following oral administration of a single 245 mg dose of tenofovir disoproxil to HIV-1 infected patients in the fasted state, maximum tenofovir concentrations were achieved within one hour. The oral bioavailability of tenofovir from tenofovir disoproxil in fasted patients was approximately 25%.

Effect of food

The fixed-dose combination of efavirenz/emtricitabine/tenofovir disoproxil has not been evaluated in the presence of food.

Efavirenz/Emtricitabine/Tenofovir Disoproxil is recommended for administration on an empty stomach since food may increase efavirenz exposure and may lead to an increase in the frequency of adverse reactions .It is anticipated that tenofovir exposure (AUC) will be approximately 30% lower following administration of Efavirenz/Emtricitabine/Tenofovir Disoproxil on an empty stomach as compared to the individual component

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tenofovir disoproxil when taken with food .

Distribution

Efavirenz is highly bound (> 99%) to human plasma proteins, predominantly albumin. After oral administration, emtricitabine is widely distributed throughout the body.

After oral administration, tenofovir is widely distributed throughout the body.

Biotransformation

Studies in humans and in vitro studies using human liver microsomes have demonstrated that efavirenz is principally metabolised by the CYP system to hydroxylated metabolites with subsequent glucuronidation of these hydroxylated metabolites. These metabolites are essentially inactive against HIV-1. The in vitro studies suggest that CYP3A4 and CYP2B6 are the major isozymes responsible for efavirenz metabolism and that it inhibits CYP isozymes 2C9, 2C19, and 3A4. In in vitro studies efavirenz did not inhibit CYP2E1 and inhibited CYP2D6 and CYP1A2 only at concentrations well above those achieved clinically.

Efavirenz plasma exposure may be increased in patients with homozygous G516T genetic variant of the CYP2B6 isozyme. The clinical implications of such an association are unknown; however, the potential for an increased frequency and severity of efavirenz-associated adverse events cannot be excluded.

Efavirenz has been shown to induce CYP3A4 and CYP2B6, resulting in the induction of its own metabolism, which may be clinically relevant in some patients. Efavirenz has also been shown to induce UGT1A1. Although in vitro data suggest that efavirenz inhibits CYP2C9 and CYP2C19, there have been contradictory reports of both increased and decreased exposures to substrates of these enzymes when co-administered with efavirenz in vivo. The net effect of co-administration is not clear. There is limited metabolism of emtricitabine. The biotransformation of emtricitabine includes oxidation of the thiol moiety to form the 3'-sulphoxide diastereomers (approximately 9% of dose) and conjugation with glucuronic acid to form 2'-O-glucuronide (approximately 4% of dose). In vitro studies have determined that neither tenofovir disoproxil nor tenofovir are substrates for the CYP enzymes. Neither emtricitabine nor tenofovir inhibited in vitro drug metabolism mediated by any of the major human CYP isoforms involved in drug biotransformation. Also, emtricitabine did not inhibit uridine 5'-diphosphoglucuronyl transferase, the enzyme responsible for glucuronidation.

Elimination

Efavirenz has a relatively long terminal half-life of at least 52 hours after single doses and 40 to 55 hours after multiple doses. Approximately 14 to 34% of a radiolabelled dose of efavirenz was recovered in the urine and less than 1% of the dose was excreted in urine as unchanged efavirenz.

Following oral administration, the elimination half-life of emtricitabine is approximately 10 hours. Emtricitabine is primarily excreted by the kidneys with complete recovery of the dose achieved in urine

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(approximately 86%) and faeces (approximately 14%). Thirteen percent of the emtricitabine dose was recovered in urine as three metabolites. The systemic clearance of emtricitabine averaged 307 ml/min. Following oral administration, the elimination half-life of tenofovir is approximately 12 to 18 hours. Tenofovir is primarily excreted by the kidneys by both filtration and an active tubular transport system with approximately 70 to 80% of the dose excreted unchanged in urine following intravenous administration. The apparent clearance of tenofovir averaged approximately 307 ml/min. Renal clearance has been estimated to be approximately 210 ml/min, which is in excess of the glomerular filtration rate. This indicates that active tubular secretion is an important part of the elimination of tenofovir.

Age

Pharmacokinetic studies have not been performed with efavirenz, emtricitabine or tenofovir in elderly patients (over 65 years of age).

Gender

The pharmacokinetics of emtricitabine and tenofovir are similar in male and female patients. Limited data suggest that females may have higher exposure to efavirenz but they do not appear to be less tolerant of efavirenz.

Ethnicity

Limited data suggest that Asian and Pacific Island patients may have higher exposure to efavirenz but they do not appear to be less tolerant of efavirenz.

Paediatric population

Pharmacokinetic studies have not been performed with the fixed-dose combination of efavirenz/emtricitabine/tenofovir disoproxil in infants and children under 18 years of age

Renal impairment

The pharmacokinetics of efavirenz, emtricitabine and tenofovir disoproxil after co-administration of the separate pharmaceutical forms or as the fixed-dose combination of efavirenz/emtricitabine/tenofovir disoproxil have not been studied in HIV infected patients with renal impairment.

Efavirenz/Emtricitabine/Tenofovir Disoproxil is not recommended for patients with moderate or severe renal impairment (creatinine clearance < 50 ml/min). Patients with moderate or severe renal impairment require dose interval adjustment of emtricitabine and tenofovir disoproxil that cannot be achieved with the combination tablet

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Hepatic impairment

The pharmacokinetics of the fixed-dose combination of efavirenz/emtricitabine/tenofovir disoproxil has not been studied in HIV infected patients with hepatic impairment. Efavirenz/Emtricitabine/Tenofovir Disoproxil should be administered with caution to patients with mild hepatic impairment

Pharmacodynamics:

Pharmacotherapeutic group: Antivirals for systemic use, antivirals for treatment of HIV infections, combinations,

ATC code: J05AR06

Mechanism of action and pharmacodynamic effects

Efavirenz is an NNRTI of HIV-1. Efavirenz non-competitively inhibits HIV-1 reverse transcriptase (RT) and does not significantly inhibit human immunodeficiency virus-2 (HIV-2) RT or cellular deoxyribonucleic acid (DNA) polymerases (α , β , γ , and δ). Emtricitabine is a nucleoside analogue of cytidine. Tenofovir disoproxil is converted in vivo to tenofovir, a nucleoside monophosphate (nucleotide) analogue of adenosine monophosphate. Emtricitabine and tenofovir are phosphorylated by cellular enzymes to form emtricitabine triphosphate and tenofovir diphosphate, respectively. In vitro studies have shown that both emtricitabine and tenofovir can be fully phosphorylated when combined together in cells. Emtricitabine triphosphate and tenofovir diphosphate competitively inhibit HIV-1 reverse transcriptase, resulting in DNA chain termination.

Both emtricitabine triphosphate and tenofovir diphosphate are weak inhibitors of mammalian DNA polymerases and there was no evidence of toxicity to mitochondria in vitro and in vivo.

Cardiac Electrophysiology

The effect of Mactrivr on the QTc interval was evaluated in an open-label, positive and placebo controlled, fixed single sequence 3-period, 3-treatment crossover QT study in 58 healthy subjects enriched for CYP2B6 polymorphisms. The mean C_{max} of efavirenz in subjects with CYP2B6 *6/*6 genotype following the administration of 600 mg daily dose for 14 days was 2.25-fold the mean C_{max} observed in subjects with CYP2B6 *1/*1 genotype. A positive relationship between efavirenz concentration and QTc prolongation was observed. Based on the concentration-QTc relationship, the mean QTc prolongation and its upper bound 90% confidence interval are 8.7 ms and 11.3 ms in subjects with CYP2B6*6/*6 genotype following the administration of 600 mg daily dose for 14 days. (see Section Warnings and Precautions & Section Interaction with Other Medicaments).

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INDICATIONS AND USAGE

Mactrivr Tablet is indicated for the treatment of HIV-1 infection in adults.

DOSAGE AND ADMINISTRATION

Adults

The dose of Mactrivr is one tablet once daily taken orally on an empty stomach. Dosing at bedtime may improve the tolerability of nervous system symptoms.

Pediatrics

Mactrivr is not recommended for use in patients <18 years of age.

Renal Impairment

Because Mactrivr is a fixed-dose combination, it should not be prescribed for patients requiring dosage adjustment such as those with moderate or severe renal impairment (creatinine clearance <50 mL/min).

Rifampicin Coadministration

When Mactrivr is administered with rifampicin to patients weighing 50 kg or more, an additional 200 mg/day of efavirenz is recommended.

ROUTE OF ADMINISTRATION

Oral

CONTRAINDICATIONS

- Hypersensitivity to the active substances or to any of the excipients of the product.
- Severe hepatic impairment [Child-Pugh-Turcotte (CPT), Class C]
- Co-administration with terfenadine, astemizole, cisapride, midazolam, triazolam, pimozide, bepridil, or ergot alkaloids (for example, ergotamine, dihydroergotamine, ergonovine, and methylegonovine). Competition for cytochrome P450 (CYP) 3A4 by efavirenz could result in inhibition of metabolism and create the potential for serious and/or life-threatening adverse reactions (for example, cardiac arrhythmias, prolonged sedation or respiratory depression) .

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-Co-administration with elbasvir/grazoprevir due to the expected significant decreases in plasma concentrations of elbasvir and grazoprevir. This effect is due to induction of CYP3A4 or P-gp by efavirenz and may result in loss of therapeutic effect of elbasvir/grazoprevir.

-Co-administration with voriconazole. Efavirenz significantly decreases voriconazole plasma concentrations while voriconazole also significantly increases efavirenz plasma concentrations. Since Mactrivir is a fixed-dose combination product, the dose of efavirenz cannot be altered.

-Co-administration with herbal preparations containing St. John's wort (*Hypericum perforatum*) due to the risk of decreased plasma concentrations and reduced clinical effects of efavirenz.

Administration to patients with:

- a family history of sudden death or of congenital prolongation of the QTc interval on electrocardiograms, or with any other clinical condition known to prolong the QTc interval.
- a history of symptomatic cardiac arrhythmias or with clinically relevant bradycardia or with congestive cardiac failure accompanied by reduced left ventricle ejection fraction.
- severe disturbances of electrolyte balance e.g. hypokalemia or hypomagnesemia.

Co-administration with drugs that are known to prolong the QTc interval (proarrhythmic). These drugs include:

- antiarrhythmics of classes IA and III,
- neuroleptics, antidepressive agents,
- certain antibiotics including some agents of the following classes: macrolides, fluoroquinolones, imidazole and triazole antifungal agents,
- certain non-sedating antihistamines (terfenadine, astemizole),
- cisapride,
- flecainide,
- certain antimalarials,
- methadone

INTERACTIONS WITH OTHER MEDICAMENTS

As Mactrivir contains efavirenz, emtricitabine and tenofovir disoproxil, any interactions that have been identified with these agents individually may occur with Mactrivir. Interaction studies with these agents have only been performed in adults.

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As a fixed combination, Mactrivr should not be administered concomitantly with other medicinal products containing the components, emtricitabine or tenofovir disoproxil. Mactrivr should not be co-administered with products containing efavirenz unless needed for dose adjustment e.g. with rifampicin. Due to similarities with emtricitabine, Mactrivr should not be administered concomitantly with other cytidine analogues, such as lamivudine. Mactrivr should not be administered concomitantly with adefovir dipivoxil or with medicinal products containing tenofovir alafenamide.

Efavirenz is an *in vivo* inducer of CYP3A4, CYP2B6 and UGT1A1. Compounds that are substrates of these enzymes may have decreased plasma concentrations when co-administered with efavirenz.

Efavirenz may be an inducer of CYP2C19 and CYP2C9; however, inhibition has also been observed *in vitro* and the net effect of co-administration with substrates of these enzymes is not clear

Efavirenz exposure may be increased when given with medicinal products (for example ritonavir) or food (for example, grapefruit juice) which inhibit CYP3A4 or CYP2B6 activity. Compounds or herbal preparations (for example Ginkgo biloba extracts and St. John's wort) which induce these enzymes may give rise to decreased plasma concentrations of efavirenz. Concomitant use of St. John's wort is contraindicated. Concomitant use of Ginkgo biloba extracts is not recommended.

In vitro and clinical pharmacokinetic interaction studies have shown the potential for CYP-mediated interactions involving emtricitabine and tenofovir disoproxil with other medicinal products is low. Cannabinoid test interaction

Efavirenz does not bind to cannabinoid receptors. False-positive urine cannabinoid test results have been reported with some screening assays in uninfected and HIV infected subjects receiving efavirenz. Confirmatory testing by a more specific method such as gas chromatography/mass spectrometry is recommended in such cases.

Contraindications of concomitant use

Mactrivr must not be administered concurrently with terfenadine, astemizole, cisapride, midazolam, triazolam, pimozide, bepridil, or ergot alkaloids (for example, ergotamine, dihydroergotamine, ergonovine, and methylegonovine), since inhibition of their metabolism may lead to serious, life-threatening event.

Elbasvir/grazoprevir: Co-administration of Mactrivr with elbasvir/grazoprevir is contraindicated because it may lead to loss of virologic response to elbasvir/grazoprevir

Voriconazole: Co-administration of standard doses of efavirenz and voriconazole is contraindicated. Since Mactrivr is a fixed-dose combination product, the dose of efavirenz cannot be altered; therefore, voriconazole and Mactrivr must not be co-administered

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St. John's wort (Hypericum perforatum): Co-administration of Mactrivir and St. John's wort or herbal preparations containing St. John's wort is contraindicated. Plasma levels of efavirenz can be reduced by concomitant use of St. John's wort due to induction of drug metabolising enzymes and/or transport proteins by St. John's wort. If a patient is already taking St. John's wort, stop St. John's wort, check viral levels and if possible efavirenz levels. Efavirenz levels may increase on stopping St. John's wort. The inducing effect of St. John's wort may persist for at least 2 weeks after cessation of treatment

QT Prolonging Drugs: There is limited information available on the potential for a pharmacodynamic interaction between Mactrivir and drugs that prolong the QTc interval. QTc prolongation has been observed with the use of efavirenz (see Section Pharmacodynamics and Section Warnings and Precautions). Consider alternatives to Mactrivir when coadministered with a drug with a known risk of Torsade de Pointes

Concomitant use not recommended

Atazanavir/ritonavir: Insufficient data are available to make a dosing recommendation for atazanavir/ritonavir in combination with Mactrivir. Therefore co-administration of atazanavir/ritonavir and Mactrivir is not recommended

Didanosine: Co-administration of Mactrivir and didanosine is not recommended

Sofosbuvir/velpatasvir and sofosbuvir/velpatasvir/voxilaprevir: Co-administration of Mactrivir and sofosbuvir/velpatasvir or

sofosbuvir/velpatasvir/voxilaprevir is not recommended

Renally eliminated medicinal products: Since emtricitabine and tenofovir are primarily eliminated by the kidneys, co-administration of Mactrivir with medicinal products that reduce renal function or compete for active tubular secretion (e.g. cidofovir) may increase serum concentrations of emtricitabine, tenofovir and/or the co-administered medicinal products.

Use of Mactrivir should be avoided with concurrent or recent use of a nephrotoxic medicinal product. Some examples include, but are not limited to, aminoglycosides, amphotericin B, foscarnet, ganciclovir, pentamidine, vancomycin, cidofovir or interleukin-2

Other interactions

Interactions between Mactrivir or its individual component(s) and other medicinal products are listed in Table 1 below (increase is indicated as “↑”, decrease as “↓”, no change as “↔”, twice daily as “b.i.d.”, once daily as “q.d.” and once every 8 hours as “q8h”).

Table 1: Interactions between Mactrivir or its individual components and other medicinal products

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Medicinal product by therapeutic areas	Effects on drug levels Change in AUC, C _{max} , C _{min}	Recommendation concerning co-administration with Mactrivr (efavirenz 600 mg, emtricitabine 200 mg, tenofovir disoproxil 245 mg)
ANTI-INFECTIVES		
HIV antivirals		
Protease inhibitors		
Atazanavir/ritonavir/Tenofovir disoproxil (300 mg q.d./100 mg q.d./245 mg q.d.)	Atazanavir: AUC: ↓ C _{max} : ↓ C _{min} : ↓ Co-administration of atazanavir/ritonavir with tenofovir resulted in increased exposure to tenofovir. Higher tenofovir concentrations could potentiate tenofovir-associated adverse events,	Co-administration of atazanavir/ritonavir and Mactrivr is not recommended.
Atazanavir/ritonavir/Efavirenz (400 mg q.d./100 mg q.d./600 mg q.d., all administered with food) Atazanavir/ritonavir/Efavirenz (400 mg q.d./200 mg q.d./600 mg q.d., all administered with food)	Atazanavir (pm): AUC: ↔* C _{max} : ↑* C _{min} : ↓* Atazanavir (pm): AUC: ↔*/** C _{max} : ↔*/** C _{min} : ↑ */** (CYP3A4 induction). * When compared to atazanavir 300	

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	mg/ritonavir 100 mg q.d. in the evening without efavirenz. This decrease in atazanavir C _{min} might negatively impact the efficacy of atazanavir. ** based on historical comparison. Co-administration of efavirenz with atazanavir/ritonavir is not	
Atazanavir/ritonavir/Emtricitabine	Interaction not studied.	
Darunavir/ritonavir/Efavirenz (300 mg b.i.d.*/100 mg b.i.d./600 mg q.d.) *lower than recommended doses; similar findings are expected with recommended doses.	Darunavir: AUC: ↓ C _{min} : ↓ C _{max} : ↓ (CYP3A4 induction) Efavirenz: AUC: ↑ C _{min} : ↑ C _{max} : ↑ (CYP3A4 inhibition)	Mactrivr in combination with darunavir/ritonavir 800/100 mg once daily may result in suboptimal darunavir C _{min} . If Mactrivr is to be used in combination with darunavir/ritonavir, the darunavir/ritonavir 600/100 mg twice daily regimen should be used.
Darunavir/ritonavir/Tenofovir disoproxil (300 mg b.i.d.*/100 mg b.i.d./245 mg q.d.) *lower than recommended dose	Darunavir: AUC: ↔ C _{min} : ↔ Tenofovir: AUC: ↑ C _{min} : ↑	Darunavir/ritonavir should be used with caution in combination with Mactrivr. See ritonavir row below.
Darunavir/ritonavir/Emtricitabine	Interaction not studied. Based on the different elimination pathways, no interaction is expected.	Monitoring of renal function may be indicated, particularly in patients with underlying systemic or
Fosamprenavir/ritonavir/Efavirenz (700 mg b.i.d./100 mg b.i.d./600 mg q.d.)	No clinically significant pharmacokinetic interaction.	Mactrivr and fosamprenavir/ritonavir
Fosamprenavir/ritonavir/Emtricitabine	Interaction not studied.	can be co-administered

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Fosamprenavir/ritonavir/Tenofovir disoproxil	Interaction not studied.	dose adjustment. See ritonavir row below.
Indinavir/Efavirenz (800 mg q8h/200 mg q.d.)	Efavirenz: AUC: ↔ C_{max}: ↔ C_{min}: ↔ Indinavir: AUC: ↓ C_{min}: ↓ A similar reduction in indinavir exposures was observed when indinavir 1,000 mg q8h was given with efavirenz 600 mg q.d. (CYP3A4 induction) For co-administration of efavirenz with low-dose ritonavir in combination with a protease inhibitor, see section on ritonavir below.	Insufficient data are available to make a dosing recommendation for indinavir when dosed with Mactrivr. While the clinical significance of decreased indinavir concentrations has not been established, the magnitude of the observed pharmacokinetic interaction should be taken into consideration when choosing a regimen containing both efavirenz, a component of Mactrivr, and indinavir.
Indinavir/Emtricitabine (800 mg q8h/200 mg q.d.)	Indinavir: AUC: ↔ C_{max}: ↔ Emtricitabine: AUC: ↔ C_{max}: ↔	
Indinavir/Tenofovir disoproxil (800 mg q8h/245 mg q.d.)	Indinavir: AUC: ↔ C_{max}: ↔ Tenofovir: AUC: ↔ C_{max}: ↔	
Lopinavir/ritonavir/Tenofovir disoproxil (400 mg b.i.d./100 mg b.i.d./245 mg q.d.)	Lopinavir/Ritonavir: AUC: ↔	Insufficient data are available to make a dosing

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	$C_{\max}$: ↔ $C_{\min}$: ↔ Tenofovir: AUC: ↑ $C_{\max}$: ↔ $C_{\min}$: ↑ Higher tenofovir concentrations could potentiate tenofovir-associated adverse events, including renal disorders.	recommendation for lopinavir/ritonavir when dosed with Mactrivr. Co-administration of lopinavir/ritonavir and Mactrivr is not recommended.
Lopinavir/ritonavir soft capsules or oral solution/Efavirenz Lopinavir/ritonavir tablets/Efavirenz (400/100 mg b.i.d./600 mg q.d.) (500/125 mg b.i.d./600 mg q.d.)	Substantial decrease in lopinavir exposure, necessitating dosage adjustment of lopinavir/ritonavir. When used in combination with efavirenz and two NRTIs, 533/133 mg lopinavir/ritonavir (soft capsules) twice daily yielded similar lopinavir plasma concentrations as compared to lopinavir/ritonavir (soft capsules) 400/100 mg twice daily without efavirenz (historical data). Lopinavir concentrations: ↓ Lopinavir concentrations: similar to lopinavir/ritonavir 400/100 mg twice daily without efavirenz. Dosage adjustment of lopinavir/ritonavir is necessary when given with efavirenz. For co-administration of efavirenz with low- dose ritonavir in combination with a protease inhibitor, see section on ritonavir below.	
Lopinavir/ritonavir/Emtricitabine	Interaction not studied.	
Ritonavir/Efavirenz	Ritonavir:	Co-administration of

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(500 mg b.i.d./600 mg q.d.)	Morning AUC: ↑ Evening AUC: ↔ Morning C_{max}: ↑ Evening C_{max}: ↔ Morning C_{min}: ↑ Evening C_{min}: ↑ Efavirenz: AUC: ↑ C_{max}: ↑ C_{min}: ↑ (inhibition of CYP-mediated oxidative metabolism) When efavirenz was given with ritonavir 500 mg or 600 mg twice daily, the combination was not well tolerated (for example, dizziness, nausea, paraesthesia and elevated liver enzymes occurred). Sufficient data on the tolerability of efavirenz with low-dose ritonavir (100 mg, once or twice daily) are not available.	ritonavir at doses of 600 mg and Mactrivr is not recommended. When using Mactrivr with low-dose ritonavir, the possibility of an increase in the incidence of efavirenz-associated adverse events should be considered, due to possible pharmacodynamic interaction.
Ritonavir/Emtricitabine	Interaction not studied.	
Ritonavir/Tenofovir disoproxil	Interaction not studied.	
Saquinavir/ritonavir/Efavirenz	Interaction not studied. For co-administration of efavirenz with low-dose ritonavir in combination with a protease inhibitor, see section on ritonavir above.	Insufficient data are available to make a dosing recommendation for saquinavir/ritonavir when dosed with Mactrivr. Co-administration of saquinavir/ritonavir and Mactrivr is not recommended. Use of
Saquinavir/ritonavir/Tenofovir disoproxil	There were no clinically significant pharmacokinetic interactions when tenofovir disoproxil was co-administered with ritonavir boosted	

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Saquinavir/ritonavir/Emtricitabine	Interaction not studied.	Mactrivr in combination with saquinavir as the sole protease inhibitor is not recommended.
CCR5 antagonist		
Maraviroc/Efavirenz (100 mg b.i.d./600 mg q.d.)	Maraviroc: AUC _{12h} : ↓ C _{max} : ↓ Efavirenz concentrations not measured, no effect is expected.	Refer to the Summary of Product Characteristics for the medicinal product containing maraviroc.
Maraviroc/Tenofovir disoproxil (300 mg b.i.d./245 mg q.d.)	Maraviroc: AUC _{12h} : ↔ C _{max} : ↔ Tenofovir concentrations not measured, no effect is expected.	
Maraviroc/Emtricitabine	Interaction not studied.	
Integrase strand transfer inhibitor		
Raltegravir/Efavirenz (400 mg single dose/-)	Raltegravir: AUC: ↓ C _{12h} : ↓ C _{max} : ↓ (UGT1A1 induction)	Mactrivr and raltegravir can be co-administered without dose adjustment.
Raltegravir/Tenofovir disoproxil (400 mg b.i.d./-)	Raltegravir: AUC: ↑ C _{12h} : ↑ C _{max} : ↑ (mechanism of interaction unknown) Tenofovir: AUC: ↓ C _{12h} : ↓ C _{max} : ↓	

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Raltegravir/Emtricitabine	Interaction not studied.	
NRTIs and NNRTIs		
NRTIs/Efavirenz	Specific interaction studies have not been performed with efavirenz and NRTIs other than lamivudine, zidovudine and tenofovir disoproxil. Clinically significant interactions have not been found and would not be expected since the NRTIs are metabolised via a different route than efavirenz and would be unlikely to compete for the same metabolic enzymes and elimination pathways.	Due to the similarity between lamivudine and emtricitabine, a component of Mactrivr, Mactrivr should not be administered concomitantly with lamivudine
NNRTIs/Efavirenz	Interaction not studied.	Since use of two NNRTIs proved not beneficial in terms of efficacy and safety, co-administration of Mactrivr and another NNRTI is not recommended.
Didanosine/Tenofovir disoproxil	Co-administration of tenofovir disoproxil and didanosine results in a 40-60% increase in systemic exposure to didanosine that may increase the risk for didanosine-related adverse reactions. Rarely, pancreatitis and lactic acidosis, sometimes fatal, have been reported. Co- administration of tenofovir disoproxil and didanosine at a dose of 400 mg daily has been associated with a significant decrease in CD4 cell count, possibly due to an intracellular	Co-administration of Mactrivr and didanosine is not recommended

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	phosphorylated (i.e. active) didanosine. A decreased dosage of 250 mg didanosine co-administered with tenofovir disoproxil therapy has been associated with reports of high rates of virologic failure within several tested combinations.	
Didanosine/Efavirenz	Interaction not studied.	
Didanosine/Emtricitabine	Interaction not studied.	
Hepatitis C antivirals		
Elbasvir/Grazoprevir + Efavirenz	Elbasvir: AUC: ↓ C_{max}: ↓ (CYP3A4 or P-gp induction - effect on elbasvir) Grazoprevir: AUC: ↓ C_{max}: ↓ (CYP3A4 or P-gp induction - effect on grazoprevir) Efavirenz: AUC: ↔ C_{max}: ↔	Co-administration of Mactrivr with elbasvir/grazoprevir is contraindicated because it may lead to loss of virologic response to elbasvir/grazoprevir. This loss is due to significant decreases in elbasvir/grazoprevir plasma concentrations caused by CYP3A4 or P-gp induction. Refer to the Summary of Product Characteristics for elbasvir/grazoprevir for
Boceprevir/Efavirenz (800 mg q8h/600 mg q.d.)	Boceprevir: AUC: ↔ * C_{max}: ↔ C_{min}: ↓ Efavirenz:	Plasma trough concentrations of boceprevir were decreased when administered with efavirenz, a component of

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	AUC: ↔ C_{max}: ↔ (CYP3A induction - effect on boceprevir) *0-8 hours No effect (↔) equals a decrease in mean ratio estimate of ≤ 20% or increase in mean ratio estimate of ≤	The clinical outcome of this observed reduction of boceprevir trough concentrations has not been directly assessed.
Ledipasvir/Sofosbuvir (90 mg/400 mg q.d.) + Efavirenz/Emtricitabine/Tenofovir disoproxil (600 mg/200 mg/245 mg q.d.)	Ledipasvir: AUC: ↓ C_{max}: ↓ C_{min}: ↓ Sofosbuvir: AUC: ↔ C_{max}: ↔ GS-331007¹: AUC: ↔ C_{max}: ↔ C_{min}: ↔ Efavirenz: AUC: ↔ C_{max}: ↔ C_{min}: ↔ Emtricitabine: AUC: ↔ C_{max}: ↔ C_{min}: ↔ Tenofovir: AUC: ↑ C_{max}: ↑ C_{min}: ↑	No dose adjustment is recommended. The increased exposure of tenofovir could potentiate adverse reactions associated with tenofovir disoproxil, including renal disorders. Renal function should be closely monitored
Sofosbuvir/Velpatasvir	Sofosbuvir:	Concomitant administration

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(400 mg/100 mg q.d.) + Efavirenz/Emtricitabine/Tenofovir disoproxil (600 mg/200 mg/245 mg q.d.)	AUC: ↔ C_{max}: ↑ GS-331007¹: AUC: ↔ C_{max}: ↔ C_{min}: ↔ Velpatasvir: AUC: ↓ C_{max}: ↓ C_{min}: ↓ Efavirenz: AUC: ↔ C_{max}: ↔ C_{min}: ↔ Emtricitabine: AUC: ↔ C_{max}: ↔ C_{min}: ↔ Tenofovir: AUC: ↑ C_{max}: ↑ C_{min}: ↑	of Mactravir and sofosbuvir/velpatasvir or sofosbuvir/velpatasvir/ voxilaprevir is expected to decrease plasma concentrations of velpatasvir and voxilaprevir. Co- administration of Mactravir with sofosbuvir/velpatasvir or sofosbuvir/velpatasvir/ voxilaprevir is not recommended
Sofosbuvir/Velpatasvir/Voxilaprevir (400 mg/100 mg/100 mg q.d.) + Efavirenz/Emtricitabine/Tenofovir disoproxil (600 mg/200 mg/245 mg q.d.)	Interaction only studied with sofosbuvir/velpatasvir. <i>Expected:</i> Voxilaprevir: ↓	
Sofosbuvir (400 mg q.d.) + Efavirenz/Emtricitabine/Tenofovir disoproxil (600 mg/200 mg/245 mg q.d.)	Sofosbuvir: AUC: ↔ C_{max}: ↓ GS-331007¹: AUC: ↔	Mactravir and sofosbuvir can be co-administered without dose adjustment.

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	C_{max}: ↓ Efavirenz: AUC: ↔ C_{max}: ↔ C_{min}: ↔ Emtricitabine: AUC: ↔ C_{max}: ↔ C_{min}: ↔ Tenofovir: AUC: ↔ C_{max}: ↑ C_{min}: ↔	
Simeprevir/Efavirenz (150 mg q.d./600 mg q.d.)	Simeprevir: AUC: ↓ C_{max}: ↓ C_{min}: ↓ Efavirenz: AUC: ↔ C_{max}: ↔ C_{min}: ↔ No effect (↔) equals a decrease in mean ratio estimate of $\leq 20\%$ or increase in mean ratio estimate of $\leq 25\%$. (CYP3A4 induction)	Concomitant administration of simeprevir with efavirenz, a component of Mactrivr, resulted in significantly decreased plasma concentrations of simeprevir due to CYP3A induction by efavirenz, which may result in loss of therapeutic effect of simeprevir. Co-administration of simeprevir with Mactrivr is not recommended.
Simeprevir/Emtricitabine	Interaction not studied. Clinically significant interactions are not expected as simeprevir and emtricitabine are eliminated via different pathways.	
Simeprevir/Tenofovir disoproxil (150 mg q.d./245 mg q.d.)	Simeprevir: AUC: ↔ C_{max}: ↔	

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	C_{min}: ↔ Tenofovir: AUC: ↔ C_{max}: ↔ C_{min}: ↔ No effect (↔) equals a decrease in mean ratio estimate of $\leq 20\%$ or increase in mean ratio estimate of $\leq 25\%$.	
Antibiotics		
Clarithromycin/Efavirenz (500 mg b.i.d./400 mg q.d.)	Clarithromycin: AUC: ↓ C_{max}: ↓ Clarithromycin 14-hydroxymetabolite: AUC: ↑ C_{max}: ↑ Efavirenz: AUC: ↔ C_{max}: ↑ (CYP3A4 induction) Rash developed in 46% of uninfected volunteers receiving efavirenz and clarithromycin.	The clinical significance of these changes in clarithromycin plasma levels is not known. Alternatives to clarithromycin (e.g. azithromycin) may be considered. Other macrolide antibiotics, such as erythromycin, have not been studied in combination with Mactrivr.
Clarithromycin/Emtricitabine	Interaction not studied.	
Clarithromycin/Tenofovir disoproxil	Interaction not studied.	
Antimycobacterials		
Rifabutin/Efavirenz (300 mg q.d./600 mg q.d.)	Rifabutin: AUC: ↓ C_{max}: ↓ C_{min}: ↓ Efavirenz: AUC: ↔	The daily dose of rifabutin should be increased by 50% when given with Mactrivr. Consider doubling the rifabutin dose in regimens where

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	C _{max} : ↔ C _{min} : ↓ (CYP3A4 induction)	3 times a week in combination with Mactrvir. The clinical effect of this dose adjustment has not been adequately evaluated. Individual tolerability and virological response should be considered when
Rifabutin/Emtricitabine	Interaction not studied.	
Rifabutin/Tenofovir disoproxil	Interaction not studied.	
Rifampicin/Efavirenz (600 mg q.d./600 mg q.d.)	Efavirenz: AUC: ↓ C _{max} : ↓ C _{min} : ↓ (CYP3A4 and CYP2B6 induction)	When Mactrvir is taken with rifampicin in patients weighing 50 kg or greater, an additional 200 mg/day (800 mg total) of efavirenz may provide exposure similar to a daily efavirenz dose of 600 mg when taken without rifampicin. The clinical effect of this dose adjustment has not been adequately evaluated. Individual tolerability and virological response should be considered when making the dose adjustment. No dose adjustment of rifampicin is recommended when given
Rifampicin/Tenofovir disoproxil (600 mg q.d./245 mg q.d.)	Rifampicin: AUC: ↔ C _{max} : ↔ Tenofovir: AUC: ↔ C _{max} : ↔	
Rifampicin/Emtricitabine	Interaction not studied.	
Antifungals		
Itraconazole/Efavirenz (200 mg b.i.d./600 mg q.d.)	Itraconazole: AUC: ↓ C _{max} : ↓	Since no dose recommendation can be made for itraconazole

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	C_{min} : ↓ (decrease in itraconazole concentrations: CYP3A4 induction) Hydroxyitraconazole: AUC: ↓ C_{max} : ↓ C_{min} : ↓ Efavirenz: AUC: ↔ C_{max} : ↔ C_{min} : ↔	used with Mactrivir, an alternative antifungal treatment should be considered.
Itraconazole/Emtricitabine	Interaction not studied.	
Itraconazole/Tenofovir disoproxil	Interaction not studied.	
Posaconazole/Efavirenz (-/400 mg q.d.)	Posaconazole: AUC: ↓ C_{max} : ↓ (UDP-G induction)	
Posaconazole/Emtricitabine	Interaction not studied.	Concomitant use of posaconazole and Mactrivir should be avoided unless the benefit to the patient outweighs the risk.
Posaconazole/Tenofovir disoproxil	Interaction not studied.	
Voriconazole/Efavirenz (200 mg b.i.d./400 mg q.d.)	Voriconazole: AUC: ↓ C_{max} : ↓ Efavirenz: AUC: ↑ C_{max} : ↑ (competitive inhibition of oxidative metabolism) Co-administration of standard doses of efavirenz and voriconazole is contraindicated	Since Mactrivir is a fixed-dose combination product, the dose of efavirenz cannot be altered; therefore, voriconazole and Mactrivir must not be co-administered.
Voriconazole/Emtricitabine	Interaction not studied.	

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Voriconazole/Tenofovir disoproxil	Interaction not studied.	
Antimalarials		
Artemether/Lumefantrine/Efavirenz (20/120 mg tablet, 6 doses of 4 tablets each over 3 days/600 mg q.d.)	Artemether: AUC: ↓ C _{max} : ↓ Dihydroartemisinin (active metabolite): AUC: ↓ C _{max} : ↓ Lumefantrine: AUC: ↓ C _{max} : ↔ Efavirenz: AUC: ↓ C _{max} : ↔ (CYP3A4 induction)	Since decreased concentrations of artemether, dihydroartemisinin, or lumefantrine may result in a decrease of antimalarial efficacy, caution is recommended when Mactrivr and artemether/lumefantrine tablets are co-administered.
Artemether/Lumefantrine/Emtricitabine	Interaction not studied.	
Artemether/Lumefantrine/Tenofovir disoproxil	Interaction not studied.	
Atovaquone and proguanil hydrochloride/Efavirenz (250/100 mg single dose/600 mg q.d.)	Atovaquone: AUC: ↓ C _{max} : ↓ Proguanil: AUC: ↓ C _{max} : ↔	Concomitant administration of atovaquone/proguanil with Mactrivr should be avoided.
Atovaquone and proguanil hydrochloride/Emtricitabine	Interaction not studied.	
Atovaquone and proguanil hydrochloride/Tenofovir disoproxil	Interaction not studied.	
ANTICONVULSANTS		
Carbamazepine/Efavirenz (400 mg q.d./600 mg q.d.)	Carbamazepine: AUC: ↓	No dose recommendation can be made for the use of

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	C_{max}: ↓ C_{min}: ↓ Efavirenz: AUC: ↓ C_{max}: ↓ C_{min}: ↓ (decrease in carbamazepine concentrations: CYP3A4 induction; decrease in efavirenz concentrations: CYP3A4 and CYP2B6 induction) Co-administration of higher doses of either efavirenz or carbamazepine has not been studied.	Mactrvir with carbamazepine. An alternative anticonvulsant should be considered. Carbamazepine plasma levels should be monitored periodically.
Carbamazepine/Emtricitabine	Interaction not studied.	
Carbamazepine/Tenofovir disoproxil	Interaction not studied.	
Phenytoin, Phenobarbital, and other anticonvulsants that are substrates of CYP isozymes	Interaction not studied with efavirenz, emtricitabine, or tenofovir disoproxil. There is a potential for reduction or increase in the plasma concentrations of phenytoin, phenobarbital and other anticonvulsants that are substrates of CYP isozymes with efavirenz.	When Mactrvir is co-administered with an anticonvulsant that is a substrate of CYP isozymes, periodic monitoring of anticonvulsant levels should be conducted.
Valproic acid/Efavirenz (250 mg b.i.d./600 mg q.d.)	No clinically significant effect on efavirenz pharmacokinetics. Limited data suggest there is no clinically significant effect on valproic acid	Mactrvir and valproic acid can be co-administered without dose adjustment. Patients should be monitored for seizure control.
Valproic acid/Emtricitabine	Interaction not studied.	
Valproic acid/Tenofovir disoproxil	Interaction not studied.	
Vigabatrin/Efavirenz Gabapentin/Efavirenz	Interaction not studied. Clinically significant interactions are not expected since vigabatrin and gabapentin are	Mactrvir and vigabatrin or gabapentin can be co-administered without dose

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	exclusively eliminated unchanged in the urine and are unlikely to compete for the same metabolic enzymes and elimination pathways as efavirenz.	adjustment.
Vigabatrin/Emtricitabine Gabapentin/Emtricitabine	Interaction not studied.	
Vigabatrin/Tenofovir disoproxil Gabapentin/Tenofovir disoproxil	Interaction not studied.	
ANTICOAGULANTS		
Warfarin/Efavirenz Acenocoumarol/Efavirenz	Interaction not studied. Plasma concentrations and effects of warfarin or acenocoumarol are potentially increased or decreased by efavirenz.	Dose adjustment of warfarin or acenocoumarol may be required when co-administered with
ANTIDEPRESSANTS		
Selective Serotonin Reuptake Inhibitors (SSRIs)		
Sertraline/Efavirenz (50 mg q.d./600 mg q.d.)	Sertraline: AUC: ↓ C _{max} : ↓ C _{min} : ↓ Efavirenz: AUC: ↔ C _{max} : ↑ C _{min} : ↔ (CYP3A4 induction)	When co-administered with Mactrivr, sertraline dose increases should be guided by clinical response.
Sertraline/Emtricitabine	Interaction not studied.	
Sertraline/Tenofovir disoproxil	Interaction not studied.	
Paroxetine/Efavirenz (20 mg q.d./600 mg q.d.)	Paroxetine: AUC: ↔ C _{max} : ↔ C _{min} : ↔	Mactrivr and paroxetine can be co-administered without dose adjustment.

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	Efavirenz: AUC: ↔ C _{max} : ↔ C _{min} : ↔	
Paroxetine/Emtricitabine	Interaction not studied.	
Paroxetine/Tenofovir disoproxil	Interaction not studied.	
Fluoxetine/Efavirenz	Interaction not studied. Since fluoxetine shares a similar metabolic profile with paroxetine, i.e. a strong CYP2D6 inhibitory effect, a similar lack of interaction would be expected for fluoxetine.	Mactrivr and fluoxetine can be co-administered without dose adjustment.
Fluoxetine/Emtricitabine	Interaction not studied.	
Fluoxetine/Tenofovir disoproxil	Interaction not studied.	
Norepinephrine and dopamine reuptake inhibitor		
Bupropion/Efavirenz [150 mg single dose (sustained release)/600 mg q.d.]	Bupropion: AUC: ↓ C _{max} : ↓ Hydroxybupropion: AUC: ↔ C _{max} : ↑ (CYP2B6 induction)	Increases in bupropion dosage should be guided by clinical response, but the maximum recommended dose of bupropion should not be exceeded. No dose adjustment is necessary for efavirenz.
Bupropion/Emtricitabine	Interaction not studied.	
Bupropion/Tenofovir disoproxil	Interaction not studied.	
CARDIOVASCULAR AGENTS		
Calcium Channel Blockers		
Diltiazem/Efavirenz (240 mg q.d./600 mg q.d.)	Diltiazem: AUC: ↓ C _{max} : ↓ C _{min} : ↓	Dose adjustments of diltiazem when co-administered with Mactrivr should be guided

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	Desacetyl diltiazem: AUC: ↓ C_{max}: ↓ C_{min}: ↓ N-monodesmethyl diltiazem: AUC: ↓ C_{max}: ↓ C_{min}: ↓ Efavirenz: AUC: ↑ C_{max}: ↑ C_{min}: ↑ (CYP3A4 induction) The increase in efavirenz pharmacokinetic parameters is not considered clinically significant.	response (refer to the Summary of Product Characteristics for diltiazem).
Diltiazem/Emtricitabine	Interaction not studied.	
Diltiazem/Tenofovir disoproxil	Interaction not studied.	
Verapamil, Felodipine, Nifedipine and Nicardipine	Interaction not studied with efavirenz, emtricitabine, or tenofovir disoproxil. When efavirenz is co-administered with a calcium channel blocker that is a substrate of the CYP3A4 enzyme, there is a potential for reduction in the plasma concentrations of the calcium channel blocker.	Dose adjustments of calcium channel blockers when co-administered with Mactrivr should be guided by clinical response (refer to the Summary of Product Characteristics for the calcium channel blocker).
LIPID LOWERING MEDICINAL PRODUCTS		
HMG Co-A Reductase Inhibitors		
Atorvastatin/Efavirenz (10 mg q.d./600 mg q.d.)	Atorvastatin: AUC: ↓ C_{max}: ↓	Cholesterol levels should be periodically monitored. Dosage adjustments of

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	2-hydroxy atorvastatin: AUC: ↓ C _{max} : ↓ 4-hydroxy atorvastatin: AUC: ↓ C _{max} : ↓ Total active HMG Co-A reductase inhibitors: AUC: ↓ C _{max} : ↓	atorvastatin may be required when co-administered with Mactrivr (refer to the Summary of Product Characteristics for atorvastatin).
Atorvastatin/Emtricitabine	Interaction not studied.	Cholesterol levels should be periodically monitored. Dosage adjustments of pravastatin may be required when co-administered with Mactrivr (refer to the Summary of Product Characteristics for
Atorvastatin/Tenofovir disoproxil	Interaction not studied.	
Pravastatin/Efavirenz (40 mg q.d./600 mg q.d.)	Pravastatin: AUC: ↓ C _{max} : ↓	
Pravastatin/Emtricitabine	Interaction not studied.	Cholesterol levels should be periodically monitored. Dosage adjustments of simvastatin may be required when co-administered with Mactrivr (refer to the Summary of Product Characteristics for
Pravastatin/Tenofovir disoproxil	Interaction not studied.	
Simvastatin/Efavirenz (40 mg q.d./600 mg q.d.)	Simvastatin: AUC: ↓ C _{max} : ↓ Simvastatin acid: AUC: ↓ C _{max} : ↓ Total active HMG Co-A reductase inhibitors: AUC: ↓ C _{max} : ↓ (CYP3A4 induction)	Cholesterol levels should be periodically monitored. Dosage adjustments of simvastatin may be required when co-administered with Mactrivr (refer to the Summary of Product Characteristics for simvastatin).

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	Co-administration of efavirenz with atorvastatin, pravastatin, or simvastatin did not affect efavirenz AUC or C _{max} values.	
Simvastatin/Emtricitabine	Interaction not studied.	
Simvastatin/Tenofovir disoproxil	Interaction not studied.	
Rosuvastatin/Efavirenz	Interaction not studied. Rosuvastatin is largely excreted unchanged via the faeces, therefore interaction with efavirenz is not expected.	Mactrivr and rosuvastatin can be co-administered without dose adjustment.
Rosuvastatin/Emtricitabine	Interaction not studied.	
Rosuvastatin/Tenofovir disoproxil	Interaction not studied.	
<i>HORMONAL CONTRACEPTIVES</i>		
Oral: Ethinylloestradiol+Norgestimate/Efavirenz (0.035 mg+0.25 mg q.d./600 mg q.d.)	Ethinylloestradiol: AUC: ↔ C _{max} : ↔ C _{min} : ↓ Norelgestromin (active metabolite): AUC: ↓ C _{max} : ↓ C _{min} : ↓ Levonorgestrel (active metabolite): AUC: ↓ C _{max} : ↓ C _{min} : ↓ (induction of metabolism) Efavirenz: no clinically significant interaction. The clinical significance of these effects is not known.	A reliable method of barrier contraception must be used in addition to hormonal contraceptives
Ethinylloestradiol/Tenofovir disoproxil	Ethinylloestradiol:	

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(-/245 mg q.d.)	AUC: ↔ C _{max} : ↔ Tenofovir: AUC: ↔ C _{max} : ↔	
Norgestimate/Ethinylestradiol/ Emtricitabine	Interaction not studied.	
Injection: Depomedroxyprogesterone acetate (DMPA)/Efavirenz (150 mg IM single dose DMPA)	No significant differences in MPA pharmacokinetic parameters were found between efavirenz-containing antiretroviral therapy and no antiretroviral therapy. Similar results were found by other investigators, although the MPA plasma levels were more variable in the second study. In both studies, plasma progesterone levels for subjects receiving efavirenz and DMPA remained low consistent with suppression of ovulation.	Because of the limited information available, a reliable method of barrier contraception must be used in addition to hormonal contraceptives
DMPA/Tenofovir disoproxil	Interaction not studied.	
DMPA/Emtricitabine	Interaction not studied.	
Implant: Etonogestrel/Efavirenz	Interaction not studied. Decreased exposure of etonogestrel may be expected (CYP3A4 induction). There have been occasional post-marketing reports of contraceptive failure with etonogestrel in efavirenz-exposed patients.	A reliable method of barrier contraception must be used in addition to hormonal contraceptives
Etonogestrel/Tenofovir disoproxil	Interaction not studied.	
Etonogestrel/Emtricitabine	Interaction not studied.	
IMMUNOSUPPRESSANTS		

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Immunosuppressants metabolised by CYP3A4 (e.g. cyclosporine, tacrolimus, sirolimus)/Efavirenz	Interaction not studied. ↓ exposure of the immunosuppressant may be expected (CYP3A4 induction). These immunosuppressants are not anticipated to impact exposure of efavirenz.	Dose adjustments of the immunosuppressant may be required. Close monitoring of immunosuppressant concentrations for at least two weeks (until stable concentrations are reached) is recommended when starting or stopping treatment with Mactrivr.
Tacrolimus/Emtricitabine/Tenofovir disoproxil (0.1 mg/kg q.d./200 mg/245 mg q.d.)	Tacrolimus: AUC: ↔ C _{max} : ↔ C _{24h} : ↔ Emtricitabine: AUC: ↔ C _{max} : ↔ C _{24h} : ↔ Tenofovir disoproxil: AUC: ↔ C _{max} : ↔ C _{24h} : ↔	
OPIOIDS		
Methadone/Efavirenz (35-100 mg q.d./600 mg q.d.)	Methadone: AUC: ↓ C _{max} : ↓ (CYP3A4 induction)	Concomitant administration with Mactrivr should be avoided due to the risk for QTc prolongation
Methadone/Tenofovir disoproxil (40-110 mg q.d./245 mg q.d.)	Methadone: AUC: ↔ C _{max} : ↔ C _{min} : ↔ Tenofovir: AUC: ↔ C _{max} : ↔ C _{min} : ↔	

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Methadone/Emtricitabine	Interaction not studied.	
Buprenorphine/naloxone/Efavirenz	Buprenorphine: AUC: Norbuprenorphine: AUC: Efavirenz: No clinically significant pharmacokinetic interaction.	Despite the decrease in buprenorphine exposure, no patients exhibited withdrawal symptoms. Dose adjustment of buprenorphine may not be necessary when co-administered with Mactrivr.
Buprenorphine/naloxone/Emtricitabine	Interaction not studied.	
Buprenorphine/naloxone/Tenofovir disoproxil	Interaction not studied.	

¹ The predominant circulating metabolite of sofosbuvir.

Studies conducted with other medicinal products

There were no clinically significant pharmacokinetic interactions when efavirenz was administered with azithromycin, cetirizine, fosamprenavir/ritonavir, lorazepam, nelfinavir, zidovudine, aluminium/magnesium hydroxide antacids, famotidine or fluconazole. The potential for interactions with efavirenz and other azole antifungals, such as ketoconazole, has not been studied.

There were no clinically significant pharmacokinetic interactions when emtricitabine was administered with stavudine, zidovudine or famciclovir. There were no clinically significant pharmacokinetic interactions when tenofovir disoproxil was co-administered with emtricitabine, nelfinavir or ribavirin.

SIDE EFFECT

Tabulated list of adverse reactions

The adverse reactions are listed in Table 2 below by body system organ class, frequency and the component(s) of Mactrivr to which the adverse reactions are attributable. Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness. Frequencies are defined as very common, common, uncommon or rare

Table 2: Adverse reactions associated with Mactrivr listed by the component(s) of Mactrivr to which the adverse reactions are attributable

	Mactrivr
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	Efavirenz	Emtricitabine	Tenofovir disoproxil
<i>Blood and lymphatic system disorders:</i>			
Common		neutropenia	
Uncommon		anaemia ¹	
<i>Immune system disorders:</i>			
Common		allergic reaction	
Uncommon	hypersensitivity		
<i>Metabolism and nutrition disorders:</i>			
Very common			hypophosphataemia ²
Common	hypertriglyceridaemia ³	hyperglycaemia, hypertriglyceridaemia	
Uncommon	hypercholesterolaemia ³		hypokalaemia ²
Rare			lactic acidosis
<i>Psychiatric disorders:</i>			
Common	depression ³ , anxiety ³ , abnormal dreams ³ , insomnia ³	abnormal dreams, insomnia	
Uncommon	suicide attempt ³ , suicide ideation ³ , psychosis ³ , mania ³ , paranoia ³ , hallucination ³ , euphoric mood ³ , affect lability ³ , confusional state ³ , aggression ³ , catatonia ³		
Rare	completed suicide ^{3,4} , delusion ^{3,4} , neurosis ^{3,4}		
<i>Nervous system disorders:</i>			
Very common		headache	dizziness
Common	cerebellar coordination and	dizziness	headache

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	balance disturbances ³ , somnolence ³ , headache ³ , disturbance in attention ³ , dizziness ³		
Uncommon	convulsions ³ , amnesia ³ , thinking abnormal ³ , ataxia ³ , coordination abnormal ³ , agitation ³ , tremor		
<i>Eye disorders:</i>			
Uncommon	vision blurred		
<i>Ear and labyrinth disorders:</i>			
Uncommon	tinnitus, vertigo		
<i>Vascular disorders:</i>			
Uncommon	flushing		
<i>Gastrointestinal disorders:</i>			
Very common		diarrhoea, nausea	diarrhoea, vomiting, nausea
Common	diarrhoea, vomiting, abdominal pain, nausea	elevated amylase including elevated pancreatic amylase, elevated serum lipase, vomiting, abdominal pain,	abdominal pain, abdominal distension, flatulence
Uncommon	pancreatitis		pancreatitis
<i>Hepatobiliary disorders:</i>			
Common	elevated aspartate aminotransferase (AST), elevated alanine aminotransferase (ALT), elevated gamma- glutamyltransferase (GGT)	elevated serum AST and/or elevated serum ALT, hyperbilirubinaemia	increased transaminases
Uncommon	hepatitis acute		

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Rare	hepatic failure ^{3,4}		hepatic steatosis, hepatitis
<i>Skin and subcutaneous tissue disorders:</i>			
Very common	rash (moderate-severe & all grades) ³		rash
Common	pruritus	vesiculobullous rash, pustular rash, maculopapular rash, rash, pruritus, urticaria, skin discolouration (increased pigmentation) ¹	
Uncommon	Stevens-Johnson syndrome, erythema multiforme ³ , severe rash	angioedema ⁴	
Rare	photoallergic dermatitis		angioedema
<i>Musculoskeletal and connective tissue disorders:</i>			
Very common		elevated creatine kinase	
Uncommon			rhabdomyolysis ² , muscular weakness ²
Rare			osteomalacia (manifested as bone pain and infrequently contributing to fractures) ^{2,4} , myopathy ²
<i>Renal and urinary disorders:</i>			
Uncommon			increased creatinine, proteinuria, proximal renal tubulopathy including Fanconi syndrome
Rare			renal failure (acute and chronic), acute tubular necrosis, nephritis

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			acute interstitial nephritis) ⁴ , nephrogenic diabetes insipidus
<i>Reproductive system and breast disorders:</i>			
Uncommon	gynaecomastia		
<i>General disorders and administration site conditions:</i>			
Very common			asthenia
Common	fatigue	pain, asthenia	

¹ Anaemia was common and skin discolouration (increased pigmentation) was very common when emtricitabine was administered to paediatric patients.

² This adverse reaction may occur as a consequence of proximal renal tubulopathy. It is not considered to be causally associated with tenofovir disoproxil in the absence of this condition.

³ See below section Description of selected adverse reactions for more details.

⁴ This adverse reaction was identified through post-marketing surveillance for either efavirenz, emtricitabine or tenofovir disoproxil.

Postmarketing experiences: encephalopathy

Description of selected adverse reactions

Rash: Rashes were usually mild-to-moderate maculopapular skin eruptions that occurred within the first two weeks of initiating therapy with efavirenz. In most patients rash resolved with continuing therapy with efavirenz within one month. Mactivir can be reinitiated in patients interrupting therapy because of rash. Use of appropriate antihistamines and/or corticosteroids is recommended when Mactivir is restarted.

Psychiatric symptoms: Patients with a history of psychiatric disorders appear to be at greater risk of serious psychiatric adverse reactions listed in the efavirenz column of Table 2.

Nervous system symptoms: Nervous system symptoms are common with efavirenz, one of the components of Mactivir. They usually begin during the first one or two days of efavirenz therapy and generally resolve after the first two to four weeks. They may occur more frequently when Mactivir is taken concomitantly with meals possibly due to increased efavirenz plasma levels. Dosing at bedtime seems to improve the tolerability of these symptoms

Hepatic failure with efavirenz: Hepatic failure, including cases in patients with no pre-existing hepatic disease or other identifiable risk factors, as reported post-marketing, were sometimes characterised by a fulminant course, progressing in some cases to transplantation or death.

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Renal impairment: As Mactrivir may cause renal damage, monitoring of renal function is recommended.

Proximal renal tubulopathy generally resolved or improved after tenofovir disoproxil discontinuation. However, in some patients, declines in creatinine clearance did not completely resolve despite tenofovir disoproxil discontinuation. Patients at risk of renal impairment (such as patients with baseline renal risk factors, advanced HIV disease, or patients receiving concomitant nephrotoxic medications) are at increased risk of experiencing incomplete recovery of renal function despite tenofovir disoproxil discontinuation.

Interaction with didanosine: Co-administration of Mactrivir and didanosine is not recommended as it results in a increase in systemic exposure to didanosine that may increase the risk of didanosine-related adverse reactions.

Rarely, pancreatitis and lactic acidosis, sometimes fatal, have been reported.

Metabolic parameters: Weight and levels of blood lipids and glucose may increase during antiretroviral therapy

Immune Reactivation Syndrome: In HIV infected patients with severe immune deficiency at the time of initiation of CART, an inflammatory reaction to asymptomatic or residual opportunistic infections may arise. Autoimmune disorders (such as Graves' disease and autoimmune hepatitis) have also been reported; however, the reported time to onset is more variable and these events can occur many months after initiation of treatment

Osteonecrosis: Cases of osteonecrosis have been reported, particularly in patients with generally acknowledged risk factors, advanced HIV disease or long-term exposure to CART. The frequency of this is unknown

Paediatric population

Insufficient safety data are available for children below 18 years of age. Mactrivir is not recommended in this population

Other special populations

Elderly: Mactrivir has not been studied in patients over the age of 65. Elderly patients are more likely to have decreased hepatic or renal function, therefore caution should be exercised when treating elderly patients with Mactrivir

Patients with renal impairment: Since tenofovir disoproxil can cause renal toxicity, close monitoring of renal function is recommended in any patient with mild renal impairment treated with Mactrivir *HIV/HBV or HCV co-infected patients:* The adverse reaction profile of efavirenz, emtricitabine and tenofovir disoproxil in patients co-infected with HIV/HBV or HIV/HCV was similar to that observed in patients infected with HIV without co-infection. However, as would be expected in this patient population, elevations in AST and ALT occurred more frequently than in the general HIV infected population.

Exacerbations of hepatitis after discontinuation of treatment: In HIV infected patients co-infected with HBV,

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clinical and laboratory evidence of hepatitis may occur after discontinuation of treatment

WARNINGS & PRECAUTIONS

Co-administration with other medicinal products

-As a fixed combination, Mactrivr should not be administered concomitantly with other medicinal products containing the same active components, emtricitabine or tenofovir disoproxil. Mactrivr should not be co-administered with products containing efavirenz unless needed for dose adjustment e.g. with rifampicin. Due to similarities with emtricitabine, Mactrivr should not be administered concomitantly with other cytidine analogues, such as lamivudine. Mactrivr should not be administered concomitantly with adefovir dipivoxil or with medicinal products containing tenofovir alafenamide.

-Co-administration of Mactrivr and didanosine is not recommended since exposure to didanosine is significantly increased following co-administration with tenofovir disoproxil that may increase the risk of didanosine-related adverse reactions. Rarely, pancreatitis and lactic acidosis, sometimes fatal have been reported.

-Co-administration of Mactrivr and sofosbuvir/velpatasvir or sofosbuvir/velpatasvir/voxilaprevir is not recommended since plasma concentrations of velpatasvir and voxilaprevir are expected to decrease following co-administration with efavirenz leading to reduced therapeutic effect of sofosbuvir/velpatasvir or sofosbuvir/velpatasvir/voxilaprevir

-No data are available on the safety and efficacy of Mactrivr in combination with other antiretroviral agents.

-Concomitant use of Ginkgo biloba extracts is not recommended

Switching from a PI-based antiretroviral regimen

Currently available data indicate a trend that in patients on a PI-based antiretroviral regimen the switch to Mactrivr may lead to a reduction of the response to the therapy. These patients should be carefully monitored for rises in viral load and, since the safety profile of efavirenz differs from that of protease inhibitors, for adverse reactions.

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Opportunistic infections

Patients receiving Mactivir or any other antiretroviral therapy may continue to develop opportunistic infections and other complications of HIV infection, and therefore should remain under close clinical observation by physicians experienced in the treatment of patients with HIV associated diseases.

Transmission of HIV

While effective viral suppression with antiretroviral therapy has been proven to substantially reduce the risk of sexual transmission, a residual risk cannot be excluded. Precautions to prevent transmission should be taken in accordance with national guidelines.

Effect of food

The administration of Mactivir with food may increase efavirenz exposure and may lead to an increase in frequency of adverse reactions. It is recommended that Mactivir be taken on an empty stomach, preferably at bedtime.

Liver disease

The pharmacokinetics, safety and efficacy of Mactivir have not been established in patients with significant underlying liver disorders. Mactivir is contraindicated in patients with severe hepatic impairment and not recommended in patients with moderate hepatic impairment. Since efavirenz is principally metabolised by the CYP system, caution should be exercised in administering Mactivir to patients with mild hepatic impairment. These patients should be carefully monitored for efavirenz adverse reactions, especially nervous system symptoms. Laboratory tests should be performed to evaluate their liver disease at periodic intervals.

Patients with pre-existing liver dysfunction including chronic active hepatitis have an increased frequency of liver function abnormalities during combination antiretroviral therapy (CART) and should be monitored according to standard practice. If there is evidence of worsening liver disease or persistent elevations of serum transaminases to greater than 5 times the upper limit of the normal range, the benefit of continued therapy with Mactivir needs to be weighed against the potential risks of significant liver toxicity. In such patients, interruption or discontinuation of treatment must be considered. In patients treated with other medicinal products associated with liver toxicity, monitoring of liver enzymes is also recommended.

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Hepatic events

Post-marketing reports of hepatic failure also occurred in patients with no pre-existing hepatic disease or other identifiable risk factors. Liver enzyme monitoring should be considered for all patients independent of pre-existing hepatic dysfunction or other risk factors.

Patients with HIV and hepatitis B (HBV) or C virus (HCV) co-infection

Patients with chronic hepatitis B or C and treated with CART are at an increased risk for severe and potentially fatal hepatic adverse reactions.

Physicians should refer to current HIV treatment guidelines for the optimal management of HIV infection in patients co-infected with HBV.

In case of concomitant antiviral therapy for hepatitis B or C, please refer also to the relevant Summary of Product Characteristics for these medicinal products.

The safety and efficacy of Mactrivr have not been studied for the treatment of chronic HBV infection. Emtricitabine and tenofovir individually and in combination have shown activity against HBV in pharmacodynamic studies. Limited clinical experience suggests that emtricitabine and tenofovir disoproxil have an anti-HBV activity when used in antiretroviral combination therapy to control HIV infection. Discontinuation of Mactrivr therapy in patients co-infected with HIV and HBV may be associated with severe acute exacerbations of hepatitis. Patients co-infected with HIV and HBV who discontinue Mactrivr must be closely monitored with both clinical and laboratory follow-up for at least four months after stopping treatment with Mactrivr. If appropriate, resumption of anti-hepatitis B therapy may be warranted. In patients with advanced liver disease or cirrhosis, treatment discontinuation is not recommended since post-treatment exacerbation of hepatitis may lead to hepatic decompensation.

QTc Prolongation

QTc prolongation has been observed with the use of efavirenz (see Section Pharmacodynamics and Section Interaction with Other Medicaments). Consider alternatives to Mactrivr when coadministered with a drug with a known risk of Torsade de Pointes or when administered to patients at higher risk of Torsade de Pointes

Psychiatric symptoms

Psychiatric adverse reactions have been reported in patients treated with efavirenz. Patients with a prior history of psychiatric disorders appear to be at greater risk of serious psychiatric adverse reactions. In

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particular, severe depression was more common in those with a history of depression. There have also been post-marketing reports of severe depression, death by suicide, delusions, psychosis-like behaviour, and catatonia. Patients should be advised that if they experience symptoms such as severe depression, psychosis or suicidal ideation, they should contact their doctor immediately to assess the possibility that the symptoms may be related to the use of efavirenz, and if so, to determine whether the risk of continued therapy outweighs the benefits

Nervous system symptoms

Symptoms including, but not limited to, dizziness, insomnia, somnolence, impaired concentration and abnormal dreaming are frequently reported undesirable effects in patients receiving efavirenz 600 mg daily in clinical studies. Dizziness was also seen in clinical studies with emtricitabine and tenofovir disoproxil. Headache has been reported in clinical studies with emtricitabine. Nervous system symptoms associated with efavirenz usually begin during the first one or two days of therapy and generally resolve after the first two to four weeks. Patients should be informed that if they do occur, these common symptoms are likely to improve with continued therapy and are not predictive of subsequent onset of any of the less frequent psychiatric symptoms.

Late-onset neurotoxicity, including ataxia and encephalopathy (impaired consciousness, confusion, psychomotor slowing, psychosis, delirium), may occur months to years after beginning efavirenz therapy. Some events of late-onset neurotoxicity have occurred in patients with CYP286 genetic polymorphisms, which are associated with increased efavirenz levels despite standard dosing of Mactrivr. Patients presenting with signs and symptoms of serious neurologic adverse experiences should be evaluated promptly to assess the possibility that these events may be related to efavirenz use, and whether discontinuation of Mactrivr is warranted.

Seizures

Convulsions have been observed in patients receiving efavirenz, generally in the presence of a known medical history of seizures. Patients who are receiving concomitant anticonvulsant medicinal products primarily metabolised by the liver, such as phenytoin, carbamazepine and phenobarbital, may require periodic monitoring of plasma levels. Carbamazepine plasma concentrations were decreased when carbamazepine was co-administered with efavirenz. Caution must be taken in any patient with a history of seizures.

Renal impairment

Mactrivr is not recommended for patients with moderate or severe renal impairment (creatinine clearance < 50 ml/min). Patients with moderate or severe renal impairment require a dose adjustment of emtricitabine and tenofovir disoproxil that cannot be achieved with the combination tablet. Use of Mactrivr should be avoided

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with concurrent or recent use of a nephrotoxic medicinal product. If concomitant use of Mactrivir and nephrotoxic agents (e.g. aminoglycosides, amphotericin B, foscarnet, ganciclovir, pentamidine, vancomycin, cidofovir, interleukin-2) is unavoidable, renal function must be monitored weekly.

Cases of acute renal failure after initiation of high dose or multiple non-steroidal anti-inflammatory drugs (NSAIDs) have been reported in patients treated with tenofovir disoproxil and with risk factors for renal dysfunction. If Mactrivir is co-administered with an NSAID, renal function should be monitored adequately.

Renal failure, renal impairment, elevated creatinine, hypophosphataemia and proximal tubulopathy (including Fanconi syndrome) have been reported with the use of tenofovir disoproxil in clinical practice

It is recommended that creatinine clearance is calculated in all patients prior to initiating therapy with Mactrivir and renal function (creatinine clearance and serum phosphate) is also monitored after two to four weeks of treatment, after three months of treatment and every three to six months thereafter in patients without renal risk factors. In patients with a history of renal dysfunction or in patients who are at risk of renal dysfunction, a more frequent monitoring of renal function is required.

If serum phosphate is < 1.5 mg/dl (0.48 mmol/l) or creatinine clearance is decreased to < 50 ml/min in any patient receiving Mactrivir, renal function must be re-evaluated within one week, including measurements of blood glucose, blood potassium and urine glucose concentrations, proximal tubulopathy). Since Mactrivir is a combination product and the dosing interval of the individual components cannot be altered, treatment with Mactrivir must be interrupted in patients with confirmed creatinine clearance < 50 ml/min or decreases in serum phosphate to < 1.0 mg/dl (0.32 mmol/l). Interrupting treatment with Mactrivir should also be considered in case of progressive decline of renal function when no other cause has been identified. Where discontinuation of therapy with one of the components of Mactrivir is indicated or where dose modification is necessary, separate preparations of efavirenz, emtricitabine and tenofovir disoproxil are available.

Bone effects

Decreases in bone mineral density of spine and changes in bone biomarkers from baseline were significantly greater in the tenofovir disoproxil treatment group at 144 weeks. Decreases in bone mineral density of the hip were significantly greater in this group until 96 weeks. However, there was no increased risk of fractures or evidence for clinically relevant bone abnormalities over 144 weeks.

The most pronounced decreases in BMD were seen in patients treated with tenofovir disoproxil as part of a regimen containing a boosted protease inhibitor. Alternative treatment regimens should be considered for patients with osteoporosis that are at a high risk for fractures.

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Bone abnormalities (infrequently contributing to fractures) may be associated with proximal renal tubulopathy. If bone abnormalities are suspected then appropriate consultation should be obtained.

Skin reactions

Mild-to-moderate rash has been reported with the individual components of Mactivir. The rash associated with the efavirenz component usually resolves with continued therapy. Appropriate antihistamines and/or corticosteroids may improve tolerability and hasten the resolution of rash. Severe rash associated with blistering, moist desquamation or ulceration has been reported in less than 1% of patients treated with efavirenz. The incidence of erythema multiforme or Stevens-Johnson syndrome was approximately 0.1%. Mactivir must be discontinued in patients developing severe rash associated with blistering, desquamation, mucosal involvement or fever. Experience with efavirenz in patients who discontinued other antiretroviral agents of the NNRTI class is limited. Mactivir is not recommended for patients who have had a life-threatening cutaneous reaction (e.g., Stevens-Johnson syndrome) while taking an NNRTI.

Weight and metabolic parameters

An increase in weight and in levels of blood lipids and glucose may occur during antiretroviral therapy. Such changes may in part be linked to disease control and life style. For lipids, there is in some cases evidence for a treatment effect, while for weight gain there is no strong evidence relating this to any particular treatment. For monitoring of blood lipids and glucose reference is made to established HIV treatment guidelines. Lipid disorders should be managed as clinically appropriate.

Mitochondrial dysfunction following exposure in utero

Nucleos(t)ide analogues may impact mitochondrial function to a variable degree, which is most pronounced with stavudine, didanosine and zidovudine. There have been reports of mitochondrial dysfunction in HIV negative infants exposed in utero and/or postnatally to nucleoside analogues; these have predominantly concerned treatment with regimens containing zidovudine. The main adverse reactions reported are haematological disorders (anaemia, neutropenia) and metabolic disorders (hyperlactatemia, hyperlipasemia). These events have often been transitory. Late onset neurological disorders have been reported rarely (hypertonia, convulsion, abnormal behaviour). Whether such neurological disorders are transient or permanent is currently unknown. These findings should be considered for any child exposed in utero to nucleos(t)ide analogues, who present with severe clinical findings of unknown etiology, particularly neurologic findings. These findings do not affect current

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national recommendations to use antiretroviral therapy in pregnant women to prevent vertical transmission of HIV.

Immune Reactivation Syndrome

In HIV infected patients with severe immune deficiency at the time of institution of CART, an inflammatory reaction to asymptomatic or residual opportunistic pathogens may arise and cause serious clinical conditions, or aggravation of symptoms. Typically, such reactions have been observed within the first few weeks or months of initiation of CART. Relevant examples are cytomegalovirus retinitis, generalised and/or focal mycobacterial infections, and *Pneumocystis jirovecii* pneumonia. Any inflammatory symptoms should be evaluated and treatment instituted when necessary.

Autoimmune disorders (such as Graves' disease and autoimmune hepatitis) have also been reported to occur in the setting of immune reactivation; however, the reported time to onset is more variable and these events can occur many months after initiation of treatment.

Osteonecrosis

Although the etiology is considered to be multifactorial (including corticosteroid use, alcohol consumption, severe immunosuppression, higher body mass index), cases of osteonecrosis have been reported particularly in patients with advanced HIV disease and/or long-term exposure to CART. Patients should be advised to seek medical advice if they experience joint aches and pain, joint stiffness or difficulty in movement.

Patients with HIV-1 harbouring mutations

Mactrivr should be avoided in patients with HIV-1 harbouring the K65R, M184V/I or K103N mutation

Elderly

Mactrivr has not been studied in patients over the age of 65. Elderly patients are more likely to have decreased hepatic or renal function, therefore caution should be exercised when treating elderly patients with Mactrivr

Effects on Ability to Drive and Use Machine

No studies on the effects on the ability to drive and use machines have been performed. However, dizziness has been reported during treatment with efavirenz, emtricitabine and tenofovir disoproxil. Efavirenz may also cause impaired concentration and/or somnolence. Patients should be instructed that

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if they experience these symptoms they should avoid potentially hazardous tasks such as driving and operating machinery.

Pregnancy Category D

Efavirenz may cause fetal harm when administered during the first trimester to a pregnant woman. Pregnancy should be avoided in women receiving Mactivir. Barrier contraception must always be used in combination with other methods of contraception (e.g., oral or other hormonal contraceptives). Because of the long half-life of efavirenz, use of adequate contraceptive measures for 12 weeks after discontinuation of Mactivir is recommended. Women of childbearing potential should undergo pregnancy testing before initiation of Mactivir. If this drug is used during the first trimester of pregnancy, or if the patient becomes pregnant while taking this drug, the patient should be apprised of the potential harm to the fetus. Mactivir should be used during pregnancy only if the potential benefit justifies the potential risk to the fetus, such as in pregnant women without other therapeutic options.

Nursing Mothers

The Centers for Disease Control and Prevention recommend that HIV-1-infected mothers not breastfeed their infants to avoid risking postnatal transmission of HIV-1. It is not known whether efavirenz, tenofovir or emtricitabine are excreted in human milk. Because of both the potential for HIV-1 transmission and the potential for serious adverse reactions in nursing infants, mothers should be instructed not to breast-feed if they are receiving Mactivir..

OVERDOSAGE

If overdose occurs, the patient should be monitored for evidence of toxicity, including monitoring of vital signs and observation of the patient's clinical status; standard supportive treatment should then be applied as necessary.

Administration of activated charcoal may be used to aid removal of unabsorbed efavirenz. Hemodialysis can remove both emtricitabine and tenofovir disoproxil fumarate (refer to detailed information below), but is unlikely to significantly remove efavirenz from the blood.

Efavirenz

Increased nervous system symptoms and involuntary muscle contractions have been reported.

Emtricitabine

Limited clinical experience is available at doses higher than the therapeutic dose of emtricitabine. No severe adverse reactions were reported with single doses of emtricitabine 1200 mg.

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It is not known whether emtricitabine can be removed by peritoneal dialysis.

Tenofovir DF

Limited clinical experience at doses higher than the therapeutic dose of tenofovir disoproxil fumarate 300 mg is available. The effects of higher doses are not known.

Tenofovir DF is efficiently removed by hemodialysis with an extraction coefficient of approximately 54%.

STORAGE

Store below 30°C in a dry place. Protect from light.

SHELF LIFE: 24 Months

PRESENTATION

30 tablets in a 100 cc HDPE bottle with Polypropylene Child resistant Closure and Polypropylene Desiccant packed in a carton along with pack insert

NAME & ADDRESS OF MANUFACTURER:

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